

1 [9231/s21/11/q7]

The curve C has equation $y = \frac{x^2 + x + 9}{x + 1}$.

(a) Find the equations of the asymptotes of C . [3]

(b) Find the coordinates of the stationary points on C . [4]

(c) Sketch C , stating the coordinates of any intersections with the axes. [3]

(d) Sketch the curve with equation $y = \left| \frac{x^2 + x + 9}{x + 1} \right|$ and find the set of values of x for which $2|x^2 + x + 9| > 13|x + 1|$. [5]

The curve C has equation $y = \frac{x^2 - x - 3}{1 + x - x^2}$.

(a) Find the equations of the asymptotes of C . [2]

(b) Find the coordinates of any stationary points on C . [3]

(c) Sketch C , stating the coordinates of the intersections with the axes. [3]

(d) Sketch the curve with equation $y = \left| \frac{x^2 - x - 3}{1 + x - x^2} \right|$ and find in exact form the set of values of x for which $\left| \frac{x^2 - x - 3}{1 + x - x^2} \right| < 3$. [6]

3 [9231/w21/11/q7]

The curve C has equation $y = \frac{4x+5}{4-4x^2}$.

- (a) Find the equations of the asymptotes of C . [2]
- (b) Find the coordinates of any stationary points on C . [4]
- (c) Sketch C , stating the coordinates of the intersections with the axes. [3]
- (d) Sketch the curve with equation $y = \left| \frac{4x+5}{4-4x^2} \right|$ and find in exact form the set of values of x for which $4|4x+5| > 5|4-4x^2|$. [6]

4 [9231/w21/12/q6]

The curve C has equation $y = \frac{x^2}{x-3}$.

(a) Find the equations of the asymptotes of C . [3]

(b) Show that there is no point on C for which $0 < y < 12$. [4]

(c) Sketch C . [2]

(d) (i) Sketch the graphs of $y = \left| \frac{x^2}{x-3} \right|$ and $y = |x| - 3$ on a single diagram, stating the coordinates of the intersections with the axes. [4]

(ii) Use your sketch to find the set of values of c for which $\left| \frac{x^2}{x-3} \right| \leq |x| + c$ has no solution. [1]

5 [9231/s20/11/q1]

Let a be a positive constant.

(a) Sketch the curve with equation $y = \frac{ax}{x+7}$. [2]

(b) Sketch the curve with equation $y = \left| \frac{ax}{x+7} \right|$ and find the set of values of x for which $\left| \frac{ax}{x+7} \right| > \frac{a}{2}$. [4]

6 [9231/s20/11/q3]

The curve C has equation $y = \frac{x^2}{2x+1}$.

(a) Find the equations of the asymptotes of C . [3]

(b) Find the coordinates of the stationary points on C . [3]

(c) Sketch C . [3]

The curve C has equation $y = \frac{10+x-2x^2}{2x-3}$.

(a) Find the equations of the asymptotes of C . [3]

(b) Show that C has no turning points. [3]

(c) Sketch C , stating the coordinates of the intersections with the axes. [3]

(d) Sketch the curve with equation $y = \left| \frac{10+x-2x^2}{2x-3} \right|$ and find in exact form the set of values of x for which $\left| \frac{10+x-2x^2}{2x-3} \right| < 4$. [6]

8 [9231/w20/11/q6]

The curve C has equation $y = \frac{x^2 + x - 1}{x - 1}$.

(a) Find the equations of the asymptotes of C . [3]

(b) Show that there is no point on C for which $1 < y < 5$. [4]

(c) Find the coordinates of the intersections of C with the axes, and sketch C . [3]

(d) Sketch the curve with equation $y = \left| \frac{x^2 + x - 1}{x - 1} \right|$. [2]

9 [9231/w20/12/q6]

Let a be a positive constant.

- (a)** The curve C_1 has equation $y = \frac{x-a}{x-2a}$. [2]

Sketch C_1 .

The curve C_2 has equation $y = \left(\frac{x-a}{x-2a}\right)^2$. The curve C_3 has equation $y = \left|\frac{x-a}{x-2a}\right|$.

- (b) (i)** Find the coordinates of any stationary points of C_2 . [3]

- (ii)** Find also the coordinates of any points of intersection of C_2 and C_3 . [3]

- (c)** Sketch C_2 and C_3 on a single diagram, clearly identifying each curve. Hence find the set of values of x for which $\left(\frac{x-a}{x-2a}\right)^2 \leq \left|\frac{x-a}{x-2a}\right|$. [5]

10 [9231/s19/11/q10]

The curves C_1 and C_2 have equations

$$y = \frac{ax}{x+5} \quad \text{and} \quad y = \frac{x^2 + (a+10)x + 5a + 26}{x+5}$$

respectively, where a is a constant and $a > 2$.

- (i) Find the equations of the asymptotes of C_1 . [2]
- (ii) Find the equation of the oblique asymptote of C_2 . [2]
- (iii) Show that C_1 and C_2 do not intersect. [2]
- (iv) Find the coordinates of the stationary points of C_2 . [3]
- (v) Sketch C_1 and C_2 on a single diagram. [You do not need to calculate the coordinates of any points where C_2 crosses the axes.] [3]

11 [9231/s19/13/q6]

The curve C has equation

$$y = \frac{x^2}{kx - 1},$$

where k is a positive constant.

- (i) Obtain the equations of the asymptotes of C . [3]
- (ii) Find the coordinates of the stationary points of C . [3]
- (iii) Sketch C . [3]

12 [9231/w19/11/q4]

The line $y = 2x + 1$ is an asymptote of the curve C with equation

$$y = \frac{x^2 + 1}{ax + b}.$$

- (i) Find the values of the constants a and b . [3]
- (ii) State the equation of the other asymptote of C . [1]
- (iii) Sketch C . [Your sketch should indicate the coordinates of any points of intersection with the y -axis. You do not need to find the coordinates of any stationary points.] [3]

13 [9231/s18/11/q6]

The curve C has equation

$$y = \frac{x^2 + b}{x + b},$$

where b is a positive constant.

- (i) Find the equations of the asymptotes of C . [3]
- (ii) Show that C does not intersect the x -axis. [1]
- (iii) Justifying your answer, find the number of stationary points on C . [2]
- (iv) Sketch C . Your sketch should indicate the coordinates of any points of intersection with the y -axis. You do not need to find the coordinates of any stationary points. [3]

14 [9231/s18/13/q4]

The curve C has equation

$$y = \frac{x^2 + 7x + 6}{x - 2}.$$

- (i) Find the coordinates of the points of intersection of C with the axes. [2]
- (ii) Find the equation of each of the asymptotes of C . [3]
- (iii) Sketch C . [3]

15 [9231/w18/11/q6]

The curve C has equation

$$y = \frac{x^2 + ax - 1}{x + 1},$$

where a is constant and $a > 1$.

- (i) Find the equations of the asymptotes of C . [3]
- (ii) Show that C intersects the x -axis twice. [1]
- (iii) Justifying your answer, find the number of stationary points on C . [2]
- (iv) Sketch C , stating the coordinates of its point of intersection with the y -axis. [3]

16 [9231/w18/12/q9]

The curve C has equation

$$y = \frac{5x^2 + 5x + 1}{x^2 + x + 1}.$$

- (i) Find the equation of the asymptote of C . [2]
- (ii) Show that, for all real values of x , $-\frac{1}{3} \leq y < 5$. [4]
- (iii) Find the coordinates of any stationary points of C . [2]
- (iv) Sketch C , stating the coordinates of any intersections with the y -axis. [2]

17 [9231/s17/11/q9]

The curve C has equation $y = \frac{x^2 - 3x + 6}{1 - x}$.

- (i) Find the equations of the asymptotes of C . [3]
- (ii) Find the coordinates of the turning points of C . [3]
- (iii) Find the coordinates of any intersections with the coordinate axes. [2]
- (iv) Sketch C . [3]

18 [9231/w17/11/q9]

The curve C has equation

$$y = \frac{3x - 9}{(x - 2)(x + 1)}.$$

- (i) Find the equations of the asymptotes of C . [2]
- (ii) Show that there is no point on C for which $\frac{1}{3} < y < 3$. [4]
- (iii) Find the coordinates of the turning points of C . [3]
- (iv) Sketch C . [3]

19 [9231/s16/11/q7]

A curve C has equation $y = \frac{x^2}{x-2}$. Find the equations of the asymptotes of C . [3]

Show that there are no points on C for which $0 < y < 8$. [4]

Sketch C , giving the coordinates of the turning points. [3]

20 [9231/s16/13/q5]

The curve C has equation $y = \frac{x+2}{x^2-9}$. Show that $\frac{dy}{dx} < 0$ at all points on C . [3]

State the equations of the asymptotes of C . [2]

Sketch C , showing the coordinates of any points of intersection with the coordinate axes. [3]

21 [9231/s15/13/q10]

The curve **C** has equation $y = \frac{4x^2 - 3x}{x^2 + 1}$. Verify that the equation of **C** may be written in the form $y = -\frac{1}{2} + \frac{(3x-1)^2}{2(x^2+1)}$ and also in the form $y = \frac{9}{2} - \frac{(x+3)^2}{2(x^2+1)}$. [3]

Hence show that $-\frac{1}{2} \leq y \leq \frac{9}{2}$. [2]

Without differentiating, write down the coordinates of the turning points of **C**. [2]

State the equation of the asymptote of **C**. [1]

Sketch the graph of **C**, stating the coordinates of the intersections with the coordinate axes and the asymptote. [3]

22 [9231/w15/11/q8]

The curve C has equation $y = \frac{2x^2 + kx}{x + 1}$, where k is a constant. Find the set of values of k for which C has no stationary points. [5]

For the case $k = 4$, find the equations of the asymptotes of C and sketch C , indicating the coordinates of the points where C intersects the coordinate axes. [6]